Unit 6, Lesson 14

Situations Involving Area

Standards

Building On	**3.MD.C.7, 4.NBT.B.5**
Addressing	**4.MD.A.3, 4.NBT.B.6, 4.OA.A.3**
Building Toward	**4.MD.A.3, 4.NBT.B.6**

Instructional Routines

- Estimation Exploration

Goals

- Compare and contrast (orally) strategies for solving area problems with two- and three-digit numbers and an unknown side length.

Student Facing Learning Goals

Let's divide to find the side length of a rectangle.

Lesson Purpose

The purpose of this lesson is for students to solve division problems that involve tiling squares and finding a side length of a rectangle with a known area in a way that makes sense to them.

Narrative

In the previous section, students related multiplication of multi-digit factors to the area of rectangles. In this lesson, students encounter division and solve the problem in a way that makes sense to them. Students find a side length of a rectangle whose area is a three-digit number and one side is a one-digit number. The context involves tiling rectangles with square tiles. This activity enables students to connect the dividend to the number of tiles in the rectangle and the divisor to the number of rectangles along one side. The provided grid encourages students to partition the area (the dividend) into smaller parts, which in turn facilitates finding the unknown length (the quotient).

In the discussion, emphasize the relationship between multiplication and division, and how students make sense of division as a way to find an unknown side length by dividing (or multiplying) in parts. Students also informally encounter leftovers in an activity in this lesson. However, they will work with quotients with remainders more formally in future lessons after they make sense of dividing with base-ten representations in the next lesson.

Access for Students with Disabilities

- Representation

Access for English Learners

- MLR7

Required Materials

Materials to Gather

- Grid paper: Activity 2
- Sticky notes: Activity 2

Lesson Timeline

Warm-up	10 min
Activity 1	15 min
Activity 2	20 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

Which questions that you asked today would you rephrase to improve students' ability to make connections or to help them better consolidate what they did? How would you rephrase them?

Warm-up

10 min

Estimation Exploration: Area of a Soccer Field

Standards

Building On	**3.MD.C.7, 4.NBT.B.5**
Building Toward	**4.MD.A.3, 4.NBT.B.6**

Instructional Routines

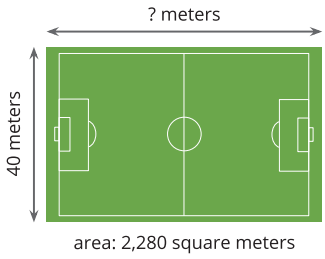
- Estimation Exploration

The purpose of this *Warm-up* is to elicit students' understandings of the relationship between the side lengths of a rectangle and its area. These understandings prepare students to reason about an unknown length or width of rectangles in the activities.

Students use their understanding about the relationship between multiplication and division, and their understanding of multiples of 10 to divide beyond 100.

Student Task Statement

Estimate: What is the length of the soccer field in meters?



area: 2,280 square meters

Record an estimate that is:

too low	about right	too high

Launch

- Groups of 2
- Display the image.
- "What is an estimate that's too high? Too low? About right?"
- 1 minute: quiet think time

Activity

- "Discuss your thinking with your partner."
- 1 minute: partner discussion
- Record responses.

Activity Synthesis

- "How did you make your estimate? How did you know it's reasonable?" (I know that the area is about 2,300 and one of the side lengths is 40. I know that $40 \times 60 = 2{,}400$ and $40 \times 50 = 2{,}000$, so the estimate is between 50 and 60, but closer to 60.)

Student Response

- Too low: 40–50 meters

- About right: 55–60 meters
- Too high: 60 meters

Consider asking:

- "Is anyone's estimate less than 40 meters? Is anyone's estimate greater than 60 meters?"
- "Based on this discussion does anyone want to revise their estimate?"

Activity 1

Elena's Mural

15 min
PLC Activity

Standards

Addressing **4.MD.A.3, 4.NBT.B.6**

In this activity, students find the length of one side of a rectangle given the length of the other side and the area of the rectangle. This work builds on what students have done in IM Grade 3, where the area was within 100 square units. In this lesson, the area is a three-digit number beyond 100.

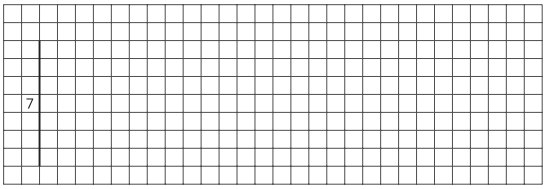
The use of tiles as a context and the presence of a grid allow students to see more concretely the relationship between a product and a factor, but the size of the product discourages students to count the tiles to find the unknown factor. Instead, students are encouraged to find multiples of the known factor or to decompose the product into parts (MP2).

Access for Students with Disabilities

Representation: Internalize Comprehension. Synthesis: Invite students to identify what information and thinking was most helpful to solve the problem. Record their responses, including mathematical language and pictures, and encourage them to use this display as a reference for the next activity.
Supports accessibility for: Conceptual Processing, Memory

Student Task Statement

Elena uses 189 square tiles to create a rectangular mural for the art club. The mural is 7 tiles wide.



1. How many tiles long is Elena's mural? Explain or show your reasoning.
2. Write one or more equations that show how you solved this problem.

Student Response

1. 27 tiles long. Sample responses:

Launch

- Groups of 2
- Explain what a mural is or show an image.

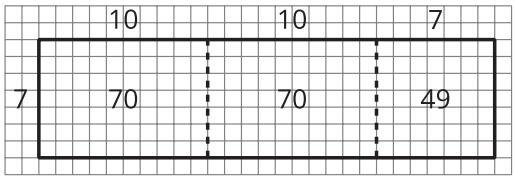
Activity

- 5 minutes: independent work time
- 2 minutes: partner discussion
- Monitor for students who use multiples of 10 and 7 and then either add or subtract multiples of 7 to reach 189.

Activity Synthesis

- Select students to share their responses and reasoning. Display their responses for all to see.

- If the length is 20 tiles, the mural would need 140 tiles, leaving 49 tiles. Seven times 7 is 49, so the length must be $20 + 7$ or 27 tiles.



- I know that 10×7 is 70. I marked off two 10×7 rectangles, which are equal to 140 tiles. $189 - 140$ is 49 and $7 \times 7 = 49$, so I marked off a 7×7 rectangle. The length of the mural is $10 + 10 + 7$, which is 27.

2. Sample responses:
   - $(7 \times 10) + (7 \times 10) + (7 \times 7) = 189$
   - $140 \div 7 = 20$, $49 \div 7 = 7$, and $20 + 7 = 27$
   - $7 \times \ell = 189$, or $\ell = 189 \div 7$

- Highlight strategies that use multiples of 10 and 7 to find the side length of the rectangle by dividing or multiplying in parts. If no students use an area diagram in their reasoning, display an example for all to see (as shown in *Student Responses*).
- Invite students to write the equations they wrote. If no division equations are included, display division equations and ask students if they could be used to answer the question. (For instance, $140 \div 7 = 20$ and $49 \div 7 = 7$, so $189 \div 7 = 27$.)

Activity 2

20 min

Tyler's Mural

Standards

Addressing **4.MD.A.3, 4.NBT.B.6, 4.OA.A.3**

This activity continues the work in the first activity. This similar context prompts students to reason about division, but the result of the division has a remainder, which students will need to interpret. After this first informal encounter with leftovers, students will interpret quotients with remainders more formally in future lessons in this unit.

After students have had some independent work time, consider a Gallery Walk of strategies.

- Post 3–4 posters around the room, each showing a likely strategy for the last problem, such as using partial products, partial quotients, pictures, or words.
- Provide some blank posters for students to show additional unique strategies.

This activity uses *MLR7 Compare and Connect*. Advances: representing, conversing

Access for English Language Learners

MLR7 Compare and Connect. Synthesis: After the Gallery Walk, lead a discussion comparing, contrasting, and connecting the different approaches. To amplify student language, and illustrate connections, follow along and point to the relevant parts of the displays as students speak.
Advances: Representing, Conversing

Required Materials

Materials to Gather

- Grid paper: Activity 2
- Sticky notes: Activity 2

Required Preparation

- If doing a Gallery Walk, create 3–4 posters to display during the activity that show or describe different strategies students are likely to use to solve the problem.

Student Task Statement

Tyler is also creating a rectangular mural for the art club. He has 197 tiles for his mural. He wants the mural to be 6 tiles wide with no gaps or overlaps between the tiles.

1. Will Tyler use all of his tiles in the mural? Explain your reasoning.
2. How many tiles long will Tyler's mural be? Show your thinking using diagrams, symbols, or other other representations.

Student Response

1. No. Sample response:
   - 30 times 6 is 180, and 3 times 6 is 18. Adding the two gives 198, not 197.
2. 32 tiles long. Sample responses:
   - $180 \div 6 = 30$, $12 \div 6 = 2$, $180 + 12 = 192$, $197 = 6 \times 32 + 5$. Tyler will not use 5 of his tiles.
   - $30 \times 6 = 180$ and $2 \times 6 = 12$, so $32 \times 6 = 192$. There will be 5 extra tiles remaining.
   - $197 = 60 + 60 + 60 + 12 + 5$

Launch

- Groups of 2
- Give access to grid paper, in case students wish to use it to create an area diagram.

Activity

MLR7 Compare and Connect

- 2–5 minutes: independent or group work
- Give each student a sticky note.
- "Make one round to visit each poster. Place your sticky note on the poster with a strategy that matches your strategy or that makes the most sense to you."
- "After your first round, make another round to visit 1–2 other posters that you didn't select. Make sense of the strategy of the poster and be prepared to explain how it is different from yours."
- 5–7 minutes: Gallery Walk

Activity Synthesis

- Discuss the results of the Gallery Walk: "Which strategy seems to be most common? The least common? Why might the strategy be the most or least common?"
- "How many tiles did Tyler use for his mural? How do you know?" (Tyler uses 192 tiles because the mural is rectangular and there are 6 rows of tiles. 192 is the greatest multiple of 6 within 197.)
- "How many tiles were not used?" ($197 - 192 = 5$. Five tiles were not used.)
- Consider asking: "How was dividing (or multiplying) in parts a helpful strategy for finding the side length?" (I

can use multiples of 10, which are easy to divide in my head. For example, the greatest multiple of 10 that is also a multiple of 6 within 197 is 180. I try to choose the greatest number so that I can keep track of the dividend. $197 - 180 = 17$. 17 is left from the dividend to divide by 6, but 17 is not a multiple of 6, so I repeat the strategy. The greatest multiple of 6 within 17 is 12, and 5 is remaining.)

Lesson Synthesis

"Today we used division to find side lengths of rectangles. For each rectangle, we knew the area and the length of one side and we used division to find the length of the other."

"What is the relationship between the side lengths and the area of a rectangle?" (The area is the product of the two side lengths.)

"How do we find the missing side length?" (Divide the area by the side length that we do know, or multiply one side length by different numbers until we find the area. We could divide or multiply in parts to find the missing side length.)

Suggested Centers

- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)
- Rolling for Fractions (3–5), Stage 2: Multiply a Fraction by a Whole Number (Supporting)

Cool-down

5 min

Sticky Notes on the Door

Standards

Addressing **4.MD.A.3, 4.NBT.B.6, 4.OA.A.3**

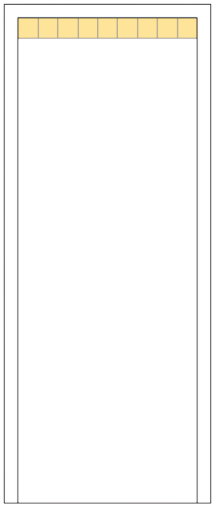
Student Task Statement

Jada's class is decorating their door with square sticky notes for their teacher. Each sticky note has a drawing or a message from a student.

The class used 234 square sticky notes to cover their classroom door completely, leaving no gaps or overlaps between the notes. It takes 9 square notes to cover the width of the door.

How many square notes does it take to cover the full height of the door? Show how you know.





Student Response

26 square notes. Sample response: I know that $9 \times 20 = 180$ and $9 \times 6 = 54$. $180 + 54 = 234$, so it takes $20 + 6$ or 26 notes to cover the height of the door.

Responding to Student Thinking

Students show they understand the problem as a division or unknown side length problem, but find a solution other than 26 square sticky notes.

Next Day Supports
After the *Warm-up* in the next lesson, pair students up to discuss their responses.

The work in this lesson builds from the area concepts developed in a prior unit.

Prior-Unit Support
Grade 3, Unit 2, Section A Concepts of Area Measurement